Experiment - 2

AIM: Setup and configure LAN (local area network) using a switch and Ethernet at your lab.

What is LAN?

A Local Area Network (LAN) refers to network that connects devices within a limited area such as office building school or home. It enables user to share resources including data printers and internet access. LAN connects devices servers and switches. Each connected device on a LAN switch can communicate directly with each other, allowing for fast and secure data transfer.

HOW TO SET UP A LAN

Step 1: plan and Design a appropriate network topology taking into account network requirement and equipment location.

Step 2: you can take 4 computers a switch with 8, 16 or 24 ports which sufficient for networks

Step 3:- connect your computers to network switch via an Ethernet cable which is simple as plugging on end of the Ethernet cable into your computer and the end into your network switch.

Step 4:- Assign IP address to your PC's.

1) Log on the client computer as Administrators.

2) click Network and internet connections.

3) Right click local Area connection Ethernet > go to properties -> select Internet protocol. (TCP / IPv4).

PC1 - IP adress . 10.1.1 subnet mask 255.0.0.0

PC2 - IP address 10.1.1.2 subnet mask 255.000.

PC3 - IP address 10.1.1.3 subnet mask 255.0.0.0

PC4 - IP - address 10.1.1.4 subnet mask 255.0.0.0

Step 5:-

1) connect your computer to the switch

2. Login in to the web interface. This open web browser. This should bring up the login page for the switch web interface.

3. Once you're logged in you will be able to configure basic setting for the switch.

4. Assign IP address 10.1.1.5 subset mask 255.0.0.0.

Step 6:- check the connectivity between switch and other machine by using ping command in command prompt of the device.

Step 7:- Select a folder -> go to properties -> click sharing tab -> share it with everyone on the same LAN

Step 8:- Try to access that shared folder from other computers of the Network.